Quantum Partition Draft: Same Φ Kernel with $\hbar, G$ Free

$a = 0.525,\ b = 0.242$ from SPARC 13/0.903 as test for $g-2$ and m_p/m_e

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Abstract

Quantum companion to the frozen galactic partition, rewritten 2026-10-07 from filed numbers (supersedes the $\hbar, G$-evolved sketch). Same kernel, same frozen shape $a = 0.525, b = 0.242$: $\Phi_q(S_{\rm hbar}, \log m/m_P) = \text{sigmoid}(a\, S_{\rm hbar} + b \log(m/m_P) + c_q)$ with cosmic $k = \ln(R_H/l_P)/16 = 8.7684$ ($R_H = c/H_0$, not fitted), per-domain $c_q = 2.90$, one global $g - 2 = C_q \Phi_q$ with $C_q = 0.001212$. Filed: electron $S = 10.118$ pred 0.00114228 vs 0.00115965 (-1.50%), muon $S = 10.726$ pred 0.00118206 vs 0.00116592 ($+1.38\%$), blind τ $+1.34\%$; proper time $\tau = 4.350$e17s includes the $30 + 220 + 600$ km/s hierarchy, no QED loops. Bound states (H/Mu/Ps/D) honestly rejected (1278%); mass ratios are identities. Status: preliminary like 13/0.903 —" two scales, one kernel, different $\hbar$. Code: `sparc-fit/eq_dsl_quantum.py`.

1 Kernel (frozen from galactic)

Galactic: $V = \sqrt{V_{\rm bar}^2 + C(L)\Phi}$, $\Phi = \text{sigmoid}(a\ell_L + b\eta_{\rm HI} + c)$, $C(L) = 2800\, L^{0.525}$; frozen blind 126/160 (7.2σ).

2 Quantum transcription (filed)

Replace $\log L \to S_{\rm hbar} = \ln(mc^2\tau/\hbar)/k$, $f_{\rm gas} \to \log(m/m_P)$, keep same a, b frozen:

$$\Phi_q = \text{sigmoid}(a\, S_{\rm hbar} + b \log(m/m_P) + c_q), \quad a = 0.525,\ b = 0.242,\ c_q = 2.90$$

$$g - 2 = C_q \Phi_q, \quad C_q = 0.001212$$

with $k = 8.7684$ from the cosmic horizon. Output $n = S/2\pi\hbar \sim$ 5e71 (e). Kill-bar test (2026-10-07): 2-loop QED gives $a_e = 0.00115964$ (matches measured 0.00115965); our pred 0.00114228 differs by 1.50% vs the 15% bar —" hint SURVIVES with $10\times$ margin. Recorded, not celebrated: margin is not confirmation. Third-prediction audit (2026-10-07): muonium HFS unreachable —" HFS needs $|\psi(0)|^2$ bound-state input absent from the kernel, and S is τ-scheme dependent (muon lifetime gives $S = 4.61$, not 10.73). No third prediction exists; branch stays a hint.

Reproducibility

`sparc-fit/eq_dsl_quantum.py` (cosmic k, c_q, C_q), GPU batched 2000. SPARC VizieR J/AJ/152/157. No new derivation claimed.

References

[1] Lelli et al., AJ 152,157 (2016).

[2] Navas, S., et al. (Particle Data Group). Phys. Rev. D 110, 030001 (2024).